Resource & Budget Plan

AI Literacy for the Modern Workforce

The solo build's direct cost was $260: about $220 of Claude subscriptions (Pro for the first two weeks, then Max 5x) plus a Google AI Pro plan for NotebookLM, with everything else on free tiers. That figure is striking but misleading in isolation: the dominant cost of any L&D project is labor, and here the practitioner's roughly 210 focused hours were uncosted, an approximate opportunity cost of about $10,500. The organizational estimate is an approximate sample that restores the full-cost picture: roughly $190,000–$258,000 for a five-person team delivering the same scope to 200 participants, showing the shape of the cost rather than a number to commit to.

Salary estimates are fully loaded (base plus benefits, taxes, and overhead, roughly 1.4×) and use market rates for mid-to-senior L&D and engineering roles in major metros (Singapore, Hong Kong, or comparable); an adopting organization would substitute its own.

SOLO DIRECT COST

$260

Out-of-pocket only. The ~210 hours of labor is an approximate opportunity cost of roughly $10,500. Everything but the AI subscriptions ran on free tiers.

ORGANIZATIONAL ESTIMATE

$190,000–$258,000

An approximate sample for a five-person team delivering the same scope to 200 participants; the shape of the cost to plan around, not a number to commit to.

ACTUAL BUILD · DIRECT COSTS

AI tools (Claude) Claude Pro for the first two weeks ($20), then Claude Max 5x for the rest of the build (two billing cycles at $100). The primary development partner for research synthesis, content authoring, code generation, and evaluation design.	**$220**
AI tools (Google AI Pro) Two months at $19.99, used for NotebookLM during research synthesis.	**$40**
Hosting (Cloudflare Pages) Free tier; static deployment with automatic SSL and CDN.	**$0**
Development tools VS Code, Node.js, Git/GitHub, all free; every dependency is an open-source npm package.	**$0**
Design tools (Claude Design) Included with the subscription; used for high-fidelity mockups and reference PDF redesigns.	**$0**

Research sources All source papers are publicly available (WEF, Anthropic Economic Index, Anthropic Interviewer).	**$0**
Authoring tools (Articulate) Deferred indefinitely; a 360 license is ~$1,399/year, not justified for a portfolio piece.	**$0**
Practitioner labor (opportunity cost) Approximate opportunity cost of roughly 210 focused hours. Not an out-of-pocket cost and not in the direct total; in any funded project this is the dominant cost.	**≈$10,500**
Total direct (out-of-pocket)	**$260**

ORGANIZATIONAL · HEADCOUNT (FTE)

L&D Manager Full project (28 wks): Program owner; splits time with other portfolio responsibilities.	**0.5 FTE**
Instructional Designer Weeks 1–24: Full-time through content development; reduces to 0.5 during platform build.	**1.0 FTE**
Full-Stack Developer Weeks 8–26: Full-time during platform build (overlaps content development); may be a contractor.	**1.0 FTE**
Subject-Matter Expert Weeks 1–18: Part-time: needs-analysis review, content accuracy, assessment validation.	**0.2 FTE**
QA Specialist Weeks 20–28: Integration testing, content QA, accessibility, browser compatibility.	**0.5 FTE**

ORGANIZATIONAL · BUDGET ESTIMATE

Labor · L&D Manager 0.5 FTE × 28 wks × ~$175–195K fully loaded	**$47,000–$53,000**
Labor · Instructional Designer 1.0 FTE × 24 wks × $90–120K fully loaded	**$42,000–$55,000**
Labor · Full-Stack Developer 1.0 FTE × 18 wks × $150–210K fully loaded (contractor may be higher)	**$52,000–$72,000**
Labor · SME 0.2 FTE × 18 wks, or external consultation	**$8,000–$15,000**
Labor · QA Specialist 0.5 FTE × 8 wks × $80–120K fully loaded	**$6,000–$9,000**
Technology + licensing Hosting, AI tools, CI/CD, testing tools	**$5,000–$10,000**

Evaluation administration Survey tools, data collection, reporting	**$5,000–$10,000**
Contingency (15%) Scope changes, extended timelines, additional review cycles	**$25,000–$34,000**
Total	**$190,000–$258,000**

Labor is **≈80%** of the total, consistent with L&D industry benchmarks. The full-stack developer is the single largest line item.

Why a custom platform, not Articulate?

Why custom code

Interactive complexity. The program includes 12 practice activities with varied interaction models: live data dashboards with filter controls, a tokenizer playground with real-time client-side tokenization, a next-token prediction simulator, a multi-step AI interaction sandbox, and a context-window failure scenario. Standard authoring tools cannot produce these natively; building them as embedded iframes would fracture the experience and block the learning-analytics tracking.

Data architecture. The platform renders six validated JSON datasets through interactive Recharts visualizations: live renderings with filters, tooltips, and guided reflections, not static images. The analytics dashboard aggregates learning events from localStorage and can export them as xAPI statements. This is beyond what an authoring tool produces without custom JavaScript, at which point the tool adds overhead rather than value.

Portfolio differentiation. For a candidate targeting L&D Manager or Senior Learning Experience Designer roles, the custom platform signals what a Storyline build does not: technical fluency, product thinking, and the ability to work with (or manage) engineering resources. The production-grade codebase is itself a reviewable artifact.

The Articulate deferral

The original architecture included a parallel Articulate build (Rise for Modules 1–2, Storyline for Modules 3–4). It was deferred indefinitely for a practical reason: the ~$1,399/year Articulate 360 license is not justified for a portfolio piece when the custom platform already demonstrates the instructional design at full fidelity.

The instructional design is platform-agnostic (only the interactive components are platform-dependent), so the content documents and component specs provide everything needed to rebuild in Articulate or any equivalent tool.

What the choice shifts

The custom approach shifts cost from licensing to labor. An Articulate build eliminates the full-stack developer line (~$52–72K) but adds licensing (~$1,399/year per seat) and constrains interaction design to what Storyline and Rise can produce. For organizations with existing Articulate licenses and Storyline-proficient designers, the authoring-tool approach is more cost-effective; for organizations that need this level of interactive complexity, or that have full-stack engineering available, the custom platform produces a superior learning experience at comparable or lower total cost.

Where the money goes

Labor accounts for about 80% of the organizational total, consistent with L&D benchmarks. The full-stack developer is the single largest line item, because the custom platform is the most labor-intensive deliverable. An organization using an off-the-shelf LMS or authoring tool would eliminate that line, and lose the interactive components (data dashboards, tokenizer playground, AI sandbox) that differentiate the program.

© Michael Ritchot. Code licensed under the MIT License. Instructional content licensed under CC BY-NC-SA 4.0.